

Samuel Tanaka Sibarani

Computer Engineering Student & Intermediate Cybersecurity Specialist

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(Remote Available) | 🌐 Open to Work - Seeking Internship Opportunities

Professional Summary

Passionate Computer Engineering student at Universitas Indonesia with intermediate-level expertise in cybersecurity, artificial intelligence, and full-stack development. Proven track record developing enterprise-grade security platforms, AI-powered systems, and production-ready applications. Certified Ethical Hacker with 15+ professional certifications and hands-on experience in penetration testing, network security, and vulnerability research. Seeking challenging internship opportunities to apply advanced technical skills and contribute to innovative technology solutions.

Education

Universitas Indonesia Jakarta, Indonesia
B.Eng. in Computer Engineering (Expected Graduation: September 2026) 09/2022 - 09/2026
Relevant Coursework: Computer Networks, Operating Systems, Database Systems, Cryptography, Digital Forensics, Machine Learning, Software Engineering

Major Projects Portfolio

NetLab V2 - Universal Parametric Cyber Range

Lead Architect & Developer | 2025

- Designed and implemented enterprise-grade cybersecurity training platform with dual-container architecture
- Created comprehensive 17-VM enterprise topology with DMZ, Corporate, and Security VLAN segmentation
- Developed web-based management dashboard with real-time network monitoring and VNC console access
- Built 5 specialized training scenarios: SOC training, penetration testing, IoT security, and incident response
- Achieved complete host system isolation with zero contamination through advanced containerization

Technologies: Docker, QEMU/KVM, Alpine/Kali/Ubuntu Linux, Windows Subsystem for Linux (WSL), Windows Subsystem for Android (WSA), Bridge Networking, Python, JavaScript, REST APIs

Impact: Complete cybersecurity training environment deployable in under 15 minutes

Banking Security Platform - AI Fraud Detection

Security Engineer & ML Developer | 2024

- Developed comprehensive fraud detection system using advanced Isolation Forest algorithm
- Implemented real-time transaction monitoring with 95%+ accuracy and customizable detection rules
- Created multi-format reporting system (PDF/CSV/Excel) with statistical analysis and trend visualization
- Built robust desktop application with modern GUI framework and efficient database management

Technologies: Python, PySide6, SQLite, scikit-learn, ReportLab, pandas

Impact: Production-ready fraud detection platform with comprehensive risk assessment capabilities

NetGuard ML - Ensemble Network Security

AI Security Researcher | 2024

- Architected ensemble machine learning framework combining multiple detection algorithms:
 - Isolation Forest for unsupervised anomaly detection
 - LSTM networks for temporal attack pattern recognition
 - Random Forest for feature-based threat classification
 - DBSCAN for attack campaign clustering analysis
- Simulated advanced persistent threats, IoT botnets, and corporate breach scenarios
- Achieved 98% detection rate with less than 0.5% false positive rate

Technologies: Python, scikit-learn, NetworkX, Matplotlib, TensorFlow

Impact: Advanced threat detection system with comprehensive attack scenario simulation

Portfolio Website - Modern Web Development

Full-Stack Developer | 2025

- Built responsive portfolio website with advanced React architecture and dynamic content management
- Implemented server-side rendering, SEO optimization, and interactive UI components
- Created comprehensive project showcase with live project demonstrations and technical documentation

Technologies: Next.js, React, TypeScript, Tailwind CSS, Shadcn/ui

Impact: Professional portfolio showcasing technical expertise and project achievements

SpecialistAI System - Multi-Agent Architecture

AI Systems Developer | 2025 - Uninished

- Designed modular multi-agent AI system with intelligent task routing and specialist coordination

- Implemented 4-bit quantization for efficient deployment in resource-constrained environments
- Built clean API interfaces with comprehensive error handling and performance optimization

Technologies: PyTorch, Python, Transformers, Phi-2, Google Colab

Impact: Efficient multi-agent AI system with minimal dependencies and optimized performance

Professional Certifications & Training

Cybersecurity Certifications 2024-2025

- 1 **Ethical Hacker - Cisco**
Cisco Networking Academy | Penetration Testing, Vulnerability Assessment, Security Analysis, Ethical Hacking
- 2 **CCNA: Enterprise Networking, Security, and Automation**
Cisco Networking Academy | Enterprise Networking, Network Security, VLAN Management, Network Automation
- 3 **CCNA: Introduction to Networks**
Cisco Networking Academy | Network Fundamentals, TCP/IP, Routing & Switching, Network Protocols

AI/ML & Database Certifications 2024-2025

- 4 **MongoDB Professional Developer Certification Series** (7 Certifications)
MongoDB University
 - **AI-Powered Search** - Vector Search, AI Integration
 - **RAG Applications** - Retrieval-Augmented Generation
 - **Advanced Schema Design** - Performance Optimization
 - **Schema Design Patterns** - Best Practices
 - **Schema Optimization** - Query Optimization
 - **CRUD Operations** - Database Operations
 - **Relational to Document Model** - Data Migration
- 5 **IBM Design Thinking Practitioner**
IBM | User Experience Design, Innovation Processes, Design Leadership

- 6 **IBM Design Thinking Co-Creator**
IBM | Collaborative Design, Team Innovation, Design Co-Creation

Specialized Security Workshops 2024-2025

- Hacktrace Ranges x Spentera, CEP-CCIT FTUI*
- **Cybersecurity Fundamentals** - Security basics, threat analysis, defense strategies
 - **Penetration Testing Basic** - Practical skills for security enthusiasts and ethical hacking
 - **Ghidra Reverse Engineering** - Binary analysis, malware research, and reverse engineering
 - **Wireless Hacking Lab** - Wi-Fi security, wireless network exploitation, and security risks

Technical Skills & Expertise

Programming Languages

- **Expert Level:** Python (Primary), C/C++
- **Proficient:** JavaScript, TypeScript, C#, Java
- **Specialized:** CUDA, OpenCL, MATLAB, Go

Cybersecurity & Penetration Testing

- **Security Tools:** Metasploit, Burp Suite, Wireshark, Nmap, Ghidra, John the Ripper, Hashcat, ZAP
- **Network Security:** Enterprise networking, VLAN configuration, network monitoring, intrusion detection
- **Vulnerability Assessment:** CVE research, security auditing, penetration testing methodologies
- **Digital Forensics:** Incident response, evidence collection, malware analysis, reverse engineering

AI/ML & Data Science

- **Frameworks:** TensorFlow, PyTorch, PyTorch, scikit-learn, Keras
- **Specializations:** Anomaly Detection, Ensemble Methods, Deep Learning, Computer Vision
- **Data Processing:** pandas, NumPy, SciPy, Matplotlib
- **Security AI:** Threat detection, fraud analysis, behavioral analytics, adversarial ML

Full-Stack Development

- **Frontend:** React, Next.js, Tailwind CSS, Vue.js
- **Backend:** FastAPI, Express.js, Django, Flask, REST APIs, GraphQL
- **Databases:** MongoDB, PostgreSQL, SQLite, Redis

- **DevOps:**  Docker,  Git, CI/CD, Linux Administration

Infrastructure & Systems

- **Virtualization:** QEMU/KVM, VMware, Docker containerization
- **Operating Systems:** Linux (Ubuntu, Kali, Alpine), Windows Server, macOS
- **Cloud Platforms:** AWS, Azure, GCP fundamentals
- **Embedded Systems:** ESP32, Arduino, STM32

CVE Research & Security Analysis

Recent Vulnerability Research & CTF Writeups 2024-2025

Billy Madison 1.1 VulnHub Machine

- Advanced port knocking exploitation and network steganography analysis
- Multi-stage attack chain from network reconnaissance to system compromise
- Packet analysis and hidden communication channel discovery

Tiki 1 VulnHub Machine

- Web application security assessment and CMS vulnerability exploitation
- SQL injection techniques and authentication bypass methods
- Privilege escalation and post-exploitation enumeration

DarkHole 2 VulnHub Machine

- Network service enumeration and systematic vulnerability assessment
- Binary exploitation and buffer overflow analysis
- Linux privilege escalation and system hardening bypass

Comprehensive security research and technical writeups: sammtan.github.io/writeups

Professional Development & Leadership

Academic Organization Leadership 2023-2024

Expert Staff - Ikatan Mahasiswa Elektro FTUI Strategic Studies and Actions Bureau | 01/2024 - 12/2024

- Led strategic research initiatives on electronics and cybersecurity technology trends
- Coordinated educational workshops and technical training programs
- Provided intelligence analysis on emerging security threats and technology developments

Staff - Ikatan Mahasiswa Elektro FTUI Strategic Studies and Actions Bureau | 01/2023 - 12/2023

- Conducted comprehensive studies on cybersecurity and electronics industry developments
- Organized community outreach programs and technical awareness campaigns
- Contributed to policy development and strategic planning initiatives

Competition Achievements 2023-2024

- **COMPFEST 15 Indie Game Ignite** - Game development competition participant (2023)
- **HackTheBox & TryHackMe** - Active cybersecurity challenge participant

Technical Security Tools Arsenal

Network Security & Analysis

- **Monitoring:** Wireshark, tcpdump, ntopng, Security Onion, Suricata
- **Scanning:** Nmap, Masscan, Zmap, Unicornscan, Nikto
- **Wireless:** Aircrack-ng, Kismet, Reaver, Wifite

Penetration Testing & Exploitation

- **Frameworks:** Metasploit, Cobalt Strike, Empire, SET
- **Web Testing:** Burp Suite, ZAP, SQLmap, w3af
- **Post-Exploitation:** Mimikatz, BloodHound, PowerShell Empire

Reverse Engineering & Malware Analysis

- **Disassemblers:** Ghidra, IDA Pro, Radare2, Binary Ninja
- **Debuggers:** GDB, WinDbg, x64dbg, OllyDbg
- **Analysis:** YARA, Volatility, Cuckoo Sandbox

Development & Automation Tools

- **IDEs:** VSCode, PyCharm, IntelliJ IDEA, Visual Studio
- **Version Control:** Git, GitHub, GitLab, Bitbucket
- **CI/CD:** GitHub Actions, Jenkins, Docker workflows

Languages & Communication

Spoken Languages

- **Indonesian:** Native proficiency
- **English:** Professional working proficiency (Technical communication, documentation, presentations)

Technical Communication

- Comprehensive technical documentation and API reference creation
- Security research publication and vulnerability disclosure
- Workshop presentation and technical training delivery
- Cross-functional team collaboration and project coordination

Research Interests & Future Goals

Cybersecurity Research Focus

- Advanced persistent threat detection and mitigation strategies
- Machine learning applications in cybersecurity and threat intelligence
- Zero-trust architecture implementation and network security automation
- Quantum computing implications for cybersecurity and cryptography

Professional Development Goals

- Contributing to open-source security tools and frameworks
- Developing innovative solutions for emerging cybersecurity challenges
- Building expertise in cloud security and container orchestration
- Advancing research in AI-powered security analytics and automation

Availability & Remote Work

Internship Availability: Immediate (Remote preferred due to current full-time studies)

Time Zone: GMT+7 (WIB) - Jakarta, Indonesia

Work Arrangement: Experienced with remote collaboration tools and distributed team workflows

Commitment: Seeking part-time internship opportunities (20-30 hours/week) compatible with academic schedule

Complete Portfolio: sammtan.github.io • *GitHub:* github.com/sammtan • *Security Research:* sammtan.github.io/writeups

"Passionate about leveraging technology to solve complex security challenges and building innovative solutions that make a meaningful impact."